Sunny Hollow AI Front Desk — Exercise Notes

Zane Meyer · Staff AI Software Engineer, Product · brightwheel
Prototype: https://ai-front-desk-eight-woad.vercel.app
Repo: https://github.com/zanemeyer/ai-front-desk
Time spent: ~3 hours

Problem

Center admins answer the same questions all day while parents wait on voicemail. An AI Front Desk only helps if answers are **center-specific, citable, and honest about uncertainty** — and if operators can close knowledge gaps without a content-ops project.

What I built

A mobile parent chat and an operator control center for a fictional center (**Sunny Hollow Early Learning Center**).

- **Parent:** ask by text (or voice in Chrome); get handbook-grounded answers with expandable source quotes, confidence, feedback, and a director handoff.
- **Operator:** dashboard of volume / auto-answer rate / gaps; inbox of questions the system struggled with; **Answer → draft entry → publish → re-test** loop; handbook editor; sensitive-topic settings.
- **Pipeline:** deterministic hard-block for unambiguous safety/legal/dispute cases → Gemini structured JSON over the full published handbook, which also judges gray-zone sensitivity against the operator's topic list → **server-side citation verification** and yes/no polarity check → fail closed to "not in handbook" when grounding is weak → log every outcome.

Seed data covers the five prompt examples and deliberately leaves gaps (sibling discount, nap blankets, potty training, birthdays) so the improvement loop is demoable.

Three design decisions

1. **Whole-handbook context, not vectors.** At ~15 policies, retrieval adds a failure mode without value. Grounding comes from **required, verified citations**, not from chunk search. Swap in retrieval at a few hundred entries.
2. **Fail closed beats a fluent guess.** Parents in early ed are anxious; a confident wrong answer is worse than a warm escalation. Low confidence / missing citations become gaps for staff, not invented policy.
3. **The inbox authors the handbook.** Unanswered questions are the backlog. Answering one *creates* a published entry. Knowledge grows from real parent demand.

What I'd do next (out of timebox)

- Persist with Redis in production (Upstash wired; in-memory fallback for demos), staff SSO on `/admin`, SMS/brightwheel message channel, eval harness in CI, and per-center multi-tenant handbooks.

Eval

`npm run eval` — 23 golden questions asserting outcome category (answered / gap / escalation) plus a few text checks (yes/no polarity, key facts). Includes false-positive guards so "Is the registration fee refundable?" is answered rather than escalated on the word "refund." **Result: 23/23 passed.**